

Assignment 1: Linear Regression Report

Data

The data I used was from the UCI ML library. It was the air quality dataset. There was no specified target variable and 15 features. I chose CO(GT) as my target variable and the goal was to predict CO(GT) concentration from co-located sensor readings.

[Link to Dataset](#)

Pre-processing

The raw file has 9751 rows with 17 columns, 2 of which were filled with null values. 113 duplicate rows were also found and removed. As for any values that were null, they were written with a value of -200 so I replaced them with NaN and dropped them. This all shrunk the dataset to just 827 rows.

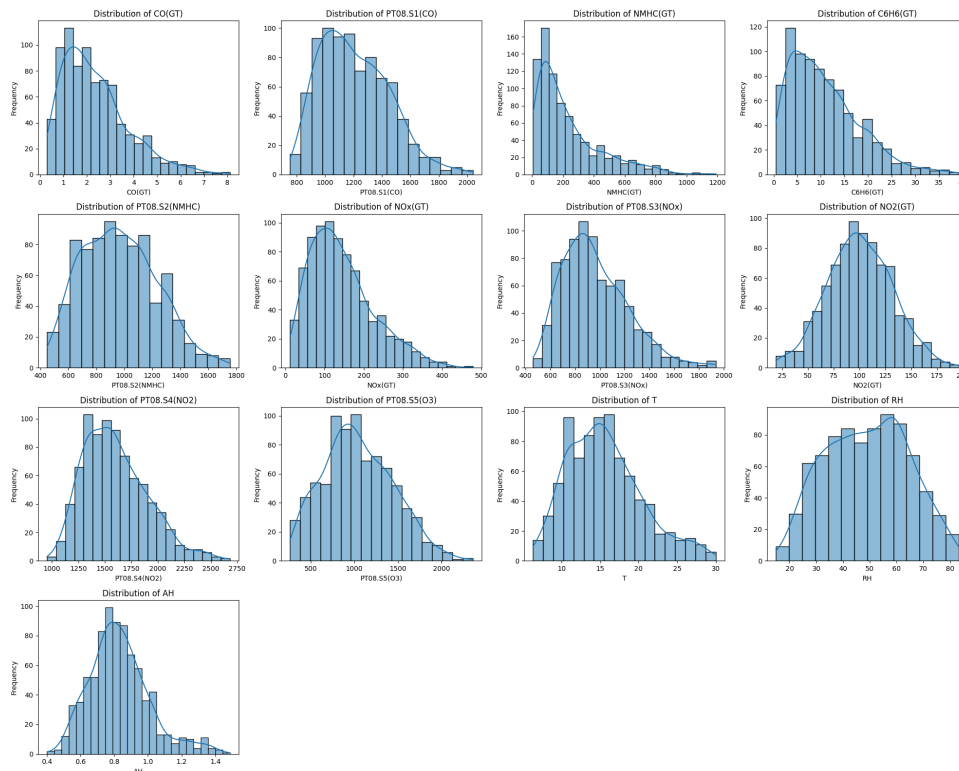


Figure 1: Distribution after cleaning

Correlation Analysis

I plotted a heatmap of my features to select the three features best correlated with my target variable (CO(GT)). The best correlated variables were C6H6(GT) = 0.97, PT08.S2(NMHC) = 0.96, NOx(GT) = 0.95. The remaining columns were excluded and these three variables were standardized before fitting SGDRegressor. I split the data 80/20 for training and testing.

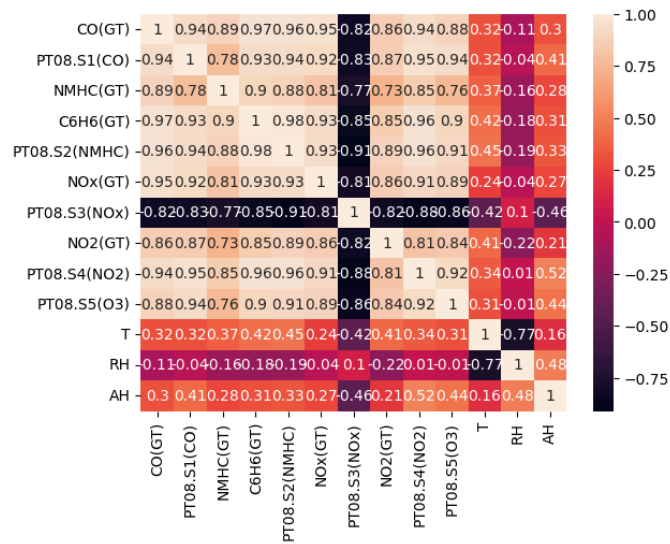


Figure 2: Heatmap to find the best correlated variables.

SGDRegressor Model

I first ran this without tuning any hyperparameters and got the following results.

	Train	Test
R^2	0.9584	0.9596
MSE	0.0795	0.0921
MAE	0.2096	0.2196

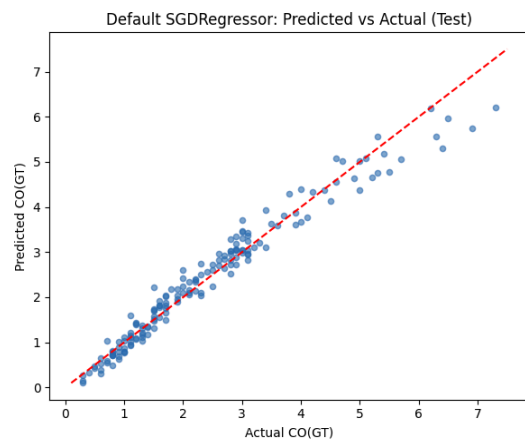


Figure 3: Regression line of Predicted vs Actual with no hyperparameter tuning

Then, I tuned the model by using a grid search which gave me 135 combinations. I chose the hyperparameter combination that gave me the best output and got the following results.

Hyperparameter	Value
Penalty	l1
Alpha	0.0001
L1 Ratio	0.5
Learning Rate	Adaptive

	Train	Test
R ²	0.9618	0.9655
MSE	0.0730	-
RMSE	-	0.3803
MAE	0.1972	0.2008

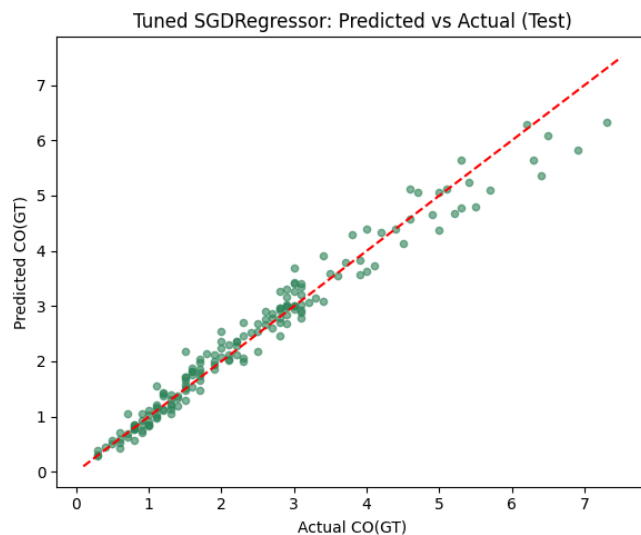


Figure 4: Regression line of Predicted vs Actual with hyperparameter tuning

As you can see, regularization slightly helped with the R² value and slightly dropped the test MAE. The winning hyperparameter combination used a very light L1 penalty with an adaptive learning rate. Since there were only three variables and not that much training data, there wasn't much need for heavy regularization.

OLS Model

	Coef	Std Err	t	p-value
const	0.4588	0.134	3.419	0.001
C6H6(GT)	0.1353	0.008	16.414	<0.001
PT08.S2(NMHC)	-0.0005	0.000	-2.137	0.033
NOx(GT)	0.0064	0.000	18.178	<0.001

I did the same thing here where I first made a model without tuning any hyperparameters and got the following.

$R^2 = 0.963$ on the training data. The F-statistic is 5628 ($p \approx 0$), so the model overall is highly statistically significant. All three predictors have $p < 0.05$, with C6H6(GT) and NOx(GT) both highly significant ($p < 0.001$) and PT08.S2(NMHC) borderline ($p = 0.033$). This is consistent with the duplicating information already carried by C6H6(GT) and NOx(GT) given the high inter-correlation among these pollutant readings. On the held-out test set: $R^2 = 0.9685$, RMSE = 0.2680, MSE = 0.0718.

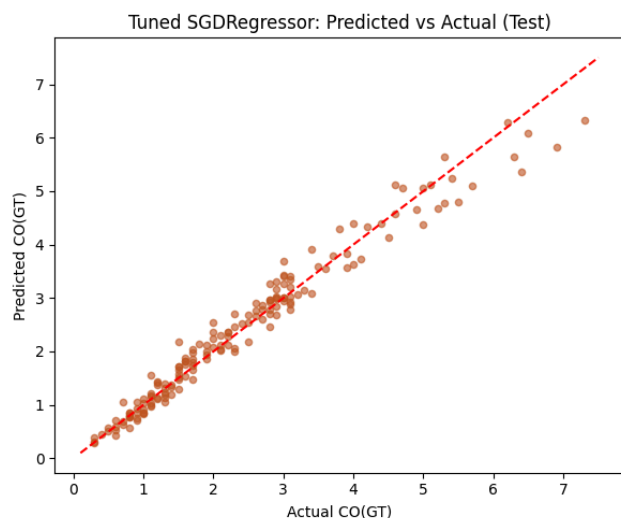


Figure 5: Unregulated OLS regression line

I did a grid search over `'alpha'` (0.0001–1.0) and `'L1_wt'` (0.0–1.0, ridge-to-lasso) and got 30 combinations that were evaluated on a test set. I got the following results.

	Test R^2	Test RMSE	Test MSE
Unregularized OLS	0.9685	0.2680	0.0718
Regularized OLS	0.9699	0.2617	0.0685

Regularization made it slightly more accurate since R^2 went up a bit and Test MSE slightly went down. However, the coefficients for the variables themselves barely moved.

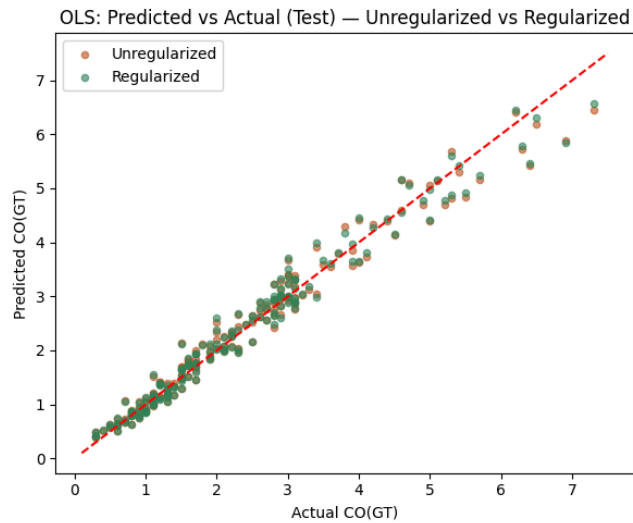


Figure 6: Comparison between regularized and unregularized residuals

Conclusion

For both types of models, whether regularized or not, there wasn't much change in the R^2 value. This wasn't very surprising since my target variable was being predicted by other pollutant concentrations measured by the same sensor at the same time so they weren't really independent variables. Also, since a lot of the data had to be dropped, it meant not as much data when it came to training causing it to only handle very little regularization. However, regularization did still show some small improvements.